title: Thruster description: The optional 10-Watt Slocum thruster: what it is, the three ways it's used (thruster-assisted yos, drift_at_depth horizontal flight, surface lens penetration), the four control modes, surface burst, energy penalty, installation and in-air checkout, post-deployment care, and field troubleshooting.

Thruster

The **thruster** is an optional propulsion module that bolts onto the tail of a G2/G3 Slocum, giving the buoyancy-driven glider an actively-powered push. It is a **high-efficiency ~10-watt** unit whose larger blades **sweep back to reduce drag when not running**, so a glider that carries one but isn't using it pays only a small flight penalty.

It buys you **speed and capability** — fighting strong currents, holding horizontal flight while drifting at depth, and "punching" through low-density surface layers — but it is **never free**: running it costs far more energy than gliding. Energy budgets must be watched carefully whenever the thruster is in use.

Source

Paraphrased from the *Slocum Glider Operators Manual* (Rev. 1, "Optional 10-Watt Thruster"), the TWR `doco/how-it-works/thruster.txt` design note, `masterdata.txt`, the Teledyne Webb Research user forum, and the UG2 community Slack. This is a condensed field reference — always defer to the current `thruster.txt` and official Teledyne documentation for your specific glider and firmware. Default values quoted here are masterdata defaults and can differ on your glider.

The energy trade-off

The thruster is a tool for getting somewhere faster or holding a position, not a default flight mode. The community consensus:

- A glider that normally burns **~3–5 coulomb A·h/day** can jump to **~10–15 A·h/day** with the thruster working — be ready for the "energy shock."